

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456547.83849 +/- 0.00094 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.035 +/- 0.062
Transit depth (Rp/Rs)^2	0.12 +/- 0.44 [%]
Orbital Inclination (inc)	83.96 +/- 2.85
Airmass coefficient 1 (a1)	37.66 +/- 26.32
Airmass coefficient 2 (a2)	-2.91 +/- 0.69
Scatter in the residuals of the lightcurve fit is	82.06 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	3.0
Transit Duration (day)	0.045 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R█ fit vs published	0.035 ± 0.062 vs 0.1646 (deviation 1.78σ)
Duration fit vs published	0.045 d vs 0.0758 d (deviation 0.94σ)
Mid-time O-C	1.4 min
Depth SNR	0.5
Residual scatter	82.06 %

Light curve

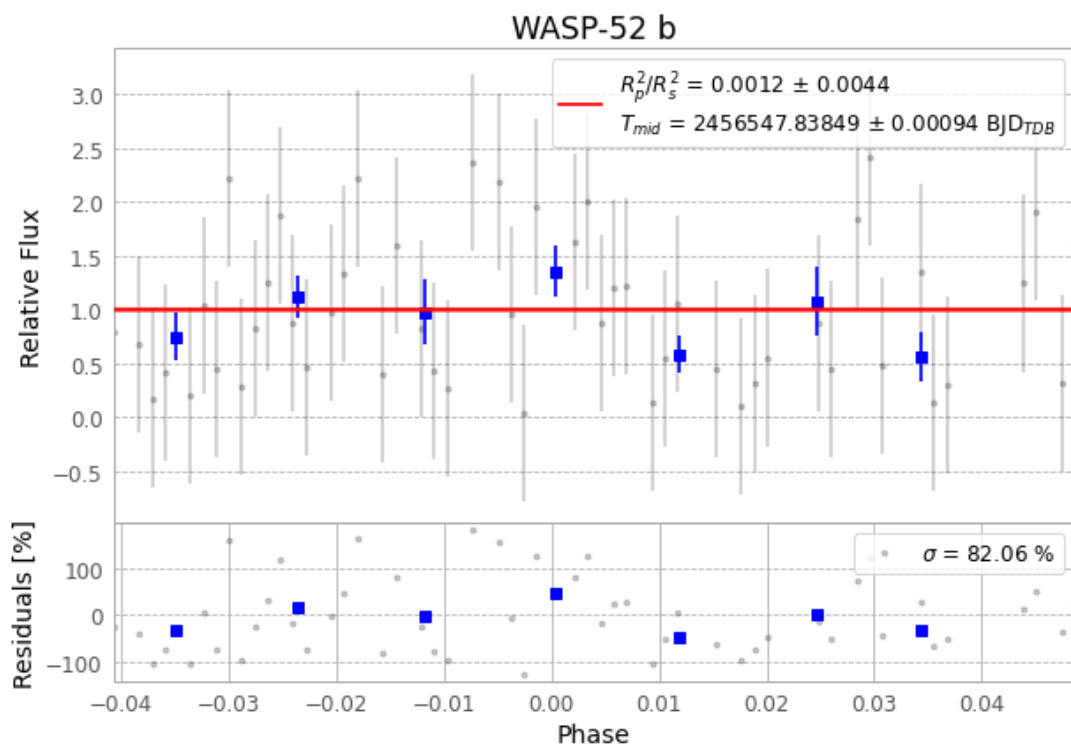


Figure 1 — FinalLightCurve_WASP-52 b_2013-09-11.png (SHA-256 add06c1159dfc4ad...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [504, 260], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5d1f7fe76126c9d2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-52130912061213.FITS → WASP-52130912100012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-130912.log (a94a3c20eff6...), FinalLightCurve_WASP-52 b_2013-09-11.png (add06c1159df...), FinalLightCurve_WASP-52 b_2013-09-11.csv (eccc18baa0a1...)

Compute resources

Wall time 117.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 01:20Z.